

SQL

Vs

MongoDB



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SQL (Structured Query Language) and **MongoDB** are both **database management systems,** but they belong to different categories and have distinct characteristics. Here's a comparison between SQL and MongoDB



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Data Model



SQL databases follow a **relational data model**. Data is structured into **tables with rows and columns**, and **relationships between tables are defined using foreign keys.**

MongoDB is a **NoSQL database** that uses a **document-oriented data model.** Data is stored in flexible, **JSON-like documents**, and relationships between data are typically represented using **embedded documents or references**.



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Schema

SQL databases have a **fixed schema where the structure of the data must be defined before storing data**. Any changes to the **schema may require altering the database schema.**





MongoDB has a **dynamic schema**, allowing documents in the **same collection to have different structures**. This flexibility makes it easier to evolve the data model as requirements change.

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Query Language



SQL databases use the **SQL query language**, which is powerful and standardized **across different database vendors.**

MongoDB uses a **query language based on JSON-like syntax.** It supports a rich set of query operations for **retrieving, updating, and deleting documents.**



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Scalability

SQL databases traditionally scale vertically by adding more resources (CPU, RAM, storage) to a **single server.**





MongoDB is designed to scale horizontally by distributing data across **multiple servers or nodes in a cluster**. This makes it easier to handle large volumes of data and traffic.

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Transactions



SQL databases support **ACID (Atomicity, Consistency, Isolation, Durability)** transactions, ensuring data integrity and consistency.

MongoDB introduced **multi-document transactions in recent versions**, providing similar transactional capabilities for operations involving **multiple documents within a single replica set or sharded cluster**.



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Complexity

SQL databases are mature and have been around for decades, with well-established principles, practices, and tools.





MongoDB and other NoSQL databases offer simplicity and flexibility but may require a different mindset for modeling data and designing applications.

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Use Cases



SQL databases are often used for applications with **complex relationships between data, transactions**, and strict consistency requirements, such as **financial systems or enterprise applications.**

MongoDB and other NoSQL databases are suitable for applications with rapidly evolving schemas, large-scale data storage, and high write throughput, such as **content management systems, IoT platforms, and real-time analytics**.



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